

AI and Its Environmental and Social Impact

Climate, Health, Governance, and Global Justice

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Before We Begin: Point at the Cloud

Everyone point — right now, with an actual finger — in the direction of **the cloud**. Nobody points at the same thing, and most people point up.

Where it actually is

It is a building. It has a street address, a parking lot, and a fence. It draws power from a substation, keeps diesel generators for when the substation fails, and has a water bill — because the heat has to go somewhere. Somebody's county council voted on it.

Commit before we go on

Name one **physical thing** that had to exist for you to send your last message. Now name a second one, further upstream. Keep going until the class runs out — and notice where you stopped.

Central Questions

- How can AI worsen environmental harms even while being advertised as a "smart" or "clean" technology?
- When does AI improve medicine and research, and when does it amplify bias, opacity, or unequal access?
- Can AI strengthen democratic governance, or does it primarily expand surveillance and bureaucratic control?
- Will AI reduce poverty and inequality, or deepen existing asymmetries of wealth, language, energy, and compute?
- What ethical principles should guide AI when the stakes are collective rather than merely individual?

The Big Idea

AI should be judged not only by whether it works, but by how it redistributes benefits, risks, resources, and decision-making power.

Connections to Previous Lectures

Lecture 1 (History): Every information revolution promised liberation while concentrating new forms of power. AI extends that pattern into energy systems, logistics, medicine, and administration.

Lecture 6 (Privacy): Medical and civic AI rely on large-scale data collection. Benefits often depend on surveillance capacities that are unevenly regulated and unequally distributed.

Lecture 7 (AI): After defining AI, the next question is not just what it *is* but what sort of world it helps build.

Lecture 8 (Work): AI affects labor directly, but also reshapes poverty, dependency, and global bargaining power.

Discussion

Is AI best understood as a set of tools, or as a new social infrastructure?

Lecture Outline

Part I: AI Is Not "Virtual"

Part II: Environment and Climate Change

Part III: Medicine and Research

Part IV: Good Governance

Part V: Poverty and Global Inequality

Part VI: Conclusion

Against the Myth of Immaterial Intelligence

Many discussions of AI treat it as if it were pure information: weightless, placeless, and detached from material constraints.

But AI depends on:

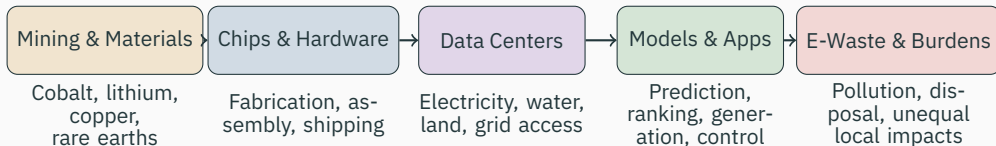
- Energy-intensive computation and cooling
- Water, land, and grid capacity for data centers
- Mineral extraction, chip fabrication, and global logistics
- Human labor for labeling, moderation, maintenance, and repair
- Public institutions that regulate, subsidize, and deploy it

crawford_atlas_ai argues that AI is best understood as a planetary-scale system of extraction, classification, and control.

Discussion Question

Why do you think people so often talk about "the cloud" as if digital systems had no physical location or environmental cost?

A Systems View of AI

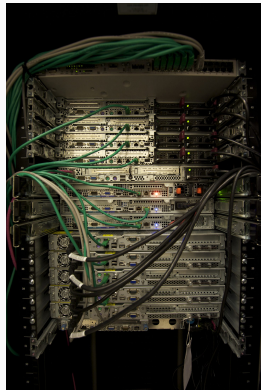


The ethical question is never just "What does the model do?" It is also: **What infrastructure makes it possible, and who pays the hidden costs?**

AI Infrastructure Begins with Physical Machines

Why This Matters

- AI runs in **server racks**, not in the cloud
- Physical infrastructure creates **energy, water, and land-use** burdens
- Costs are local even when the benefits are global
- This makes AI an environmental ethics issue, not just a software ethics issue



Servers in a rack. CC BY-SA 3.0, via Wikimedia Commons.

Why Collective Harms Are Hard to See

Many AI harms are **distributed**, **delayed**, and **offloaded**.

- A chatbot query feels trivial to the user, but scales across millions of users and large infrastructures.
- Benefits are often immediate and visible: speed, convenience, novelty.
- Costs are often indirect: emissions, water consumption, labor exploitation, grid congestion.
- The people who benefit are often not the same people who bear the costs.

Moral lesson

Individual convenience can mask collective harm. That is why environmental and social impact require institutional, not merely personal, ethics.

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AI's Environmental Footprint

Dimension	Why it matters
Electricity	Training and inference require large and growing power loads
Water	Cooling systems can place pressure on local water supplies
Materials	Chips depend on extraction, fabrication, and transport chains
Land & grids	Hyperscale facilities compete for physical space and grid capacity
Waste	Hardware turnover creates disposal and recycling burdens

The key mistake is to think of AI as only a question of software efficiency. It is also a question of **infrastructure politics**.

Current Environmental Statistics

- Data centers used roughly **415 TWh in 2024** (about **1.5%** of world electricity), rising to about **485 TWh in 2025** ([iea_energy_and_ai_2025](#); [iea_electricity_2026](#)).
- The IEA projects that roughly **doubling to ~950 TWh by 2030**, about **3%** of global electricity. **AI-specific** use grew about **50% in 2025** alone ([iea_electricity_2026](#)).
- In the **United States**, data centers account for close to **half** of all projected electricity demand growth through 2030 ([iea_electricity_2026](#)).
- More than **2,500 GW** of projects sit stalled in grid connection queues worldwide ([iea_electricity_2026](#)).

Interpretive Point

AI's footprint is not apocalyptic by itself, but it is politically significant because it compounds other pressures on already-stressed energy and water systems.

Efficiency is not the same as reduction

Technologies can become more efficient while total resource use still rises if demand expands faster than efficiency improves.

Applied to AI:

- Models become cheaper per query
- Firms deploy them in more products and at larger scales
- Total inference volume rises dramatically
- Aggregate electricity and hardware demand still grows

This is why "smaller and cheaper models" can be environmentally ambiguous: lower unit cost may increase total use.

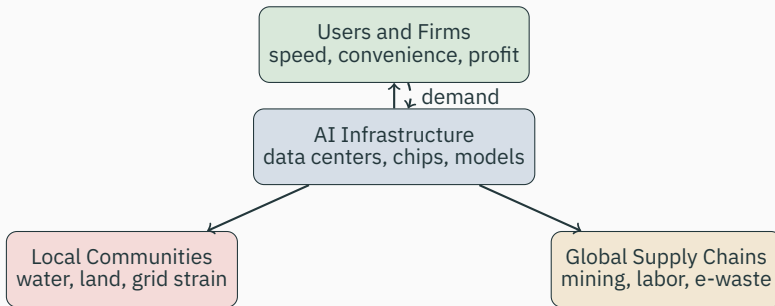
Training vs. Inference

Early public discussion focused on the cost of **training** frontier models. But recent evidence suggests that **inference** can be the larger long-run burden.

Stage	Main cost pattern	Ethical relevance
Training	Large one-time or periodic compute spikes	Concentrated in frontier firms and labs
Inference	Ongoing cost per query or use	Scales with mass adoption and everyday use
Deployment	Integration into search, office tools, medicine, government	Converts technical possibility into social impact

The environmental story is therefore not just about spectacular training runs. It is about what happens when AI becomes ordinary infrastructure.

Who Gets the Benefits and Who Bears the Burdens?



One reason AI ethics is politically difficult: the people receiving the **benefits** are often not the people absorbing the **costs**.

The skeptical view

Much "AI for climate" rhetoric is marketing. The same firms that promise optimization and sustainability are driving demand for energy, chips, and cloud infrastructure at extraordinary scale.

Skeptical questions:

- Are the climate benefits real, or merely pilot projects and demos?
- Are firms comparing AI's footprint to realistic alternatives?
- Are they disclosing full environmental costs, including hardware and water?

This objection does not prove AI cannot help with climate. It shows why environmental claims need evidence rather than branding.

AI Can Also Help with Climate

Mitigation uses

- Grid balancing and demand forecasting
- Building-efficiency optimization
- Materials discovery
- Precision agriculture and logistics

Adaptation uses

- Wildfire and flood prediction
- Disaster response coordination
- Crop-risk forecasting
- Early-warning systems for extreme events

The environmental question is therefore *comparative*: do specific uses of AI reduce enough harm to justify their material costs?

Example

If AI helps an electric grid shift demand away from peak hours, that may reduce blackouts, lower costs, and make it easier to integrate renewables. This is a stronger climate case than merely using AI to generate more content more efficiently.

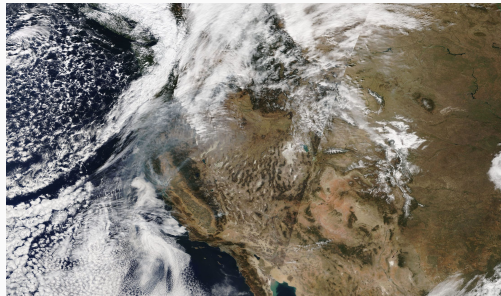
AI and Climate Adaptation: Disaster Detection from Space

Potential Benefit

- Satellite imagery and machine learning can help detect **wildfires**, smoke plumes, and changing risk conditions
- Faster detection can improve evacuation, containment, and resource allocation
- This is one of the clearest examples of AI serving a genuine public good

But

Public value depends on who has access to the tools, the data, and the response capacity.



NASA wildfire satellite image. CC BY 2.0, via Wikimedia Commons.

Thought Question: When Is "AI for Climate" Ethically Justified?

Discussion Question

Suppose an AI system consumes substantial energy but improves wildfire detection, flood planning, or grid balancing. What standard should we use to judge whether it is justified?

Possible answers:

- Net emissions reduced over time
- Lives saved or harms averted
- Benefits distributed fairly
- Full environmental costs publicly disclosed

Which of these is necessary? Which are sufficient?

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Why Medicine Looks Like AI's Best Case

AI appears especially attractive in medicine because the stakes are tangible:

- Better diagnosis and triage
- Faster image analysis
- More efficient paperwork and workflow
- Drug discovery and protein-structure prediction
- Potentially improved access in under-resourced settings

Unlike many AI use cases, medical gains can be stated in terms of **reduced suffering, fewer missed diagnoses**, and **accelerated research**.

Current Medical AI Statistics

- Stanford HAI reports that the FDA approved **223 AI-enabled medical devices in 2023**, up from just **6 in 2015** (`stanford_ai_index_2025`).
- The FDA's public list passed **1,500 AI-enabled devices** in early 2026 and is updated several times a year (`fda_ai_medical_devices_2026`). Roughly **three-quarters are radiology** devices—the adoption is real but highly concentrated.
- Read that list carefully: the FDA itself cautions it is **not a comprehensive inventory**, but a keyword-derived sample drawn from public authorization summaries. Even our regulatory statistics are constructed artifacts.
- Of those, **radiology dominates** the field: 1,104 devices, followed by cardiovascular and neurology applications.

FDA panel (lead)	Count
Radiology	1104
Cardiovascular	141
Neurology	67
Anesthesiology	27
Gastroenterology–Urology	26

AI in Research: The AlphaFold Example

A scientific success story

Protein-structure prediction was a longstanding problem in biology. **jumper_alphafold_2021** showed that deep learning could produce a major breakthrough.

Why this matters ethically:

- It shows AI can genuinely accelerate science
- Benefits may spill over into drug discovery and biomedical research
- It strengthens the case that some AI applications create broad social value

But scientific promise does not erase concerns about access, patenting, data control, and who can benefit from these advances.

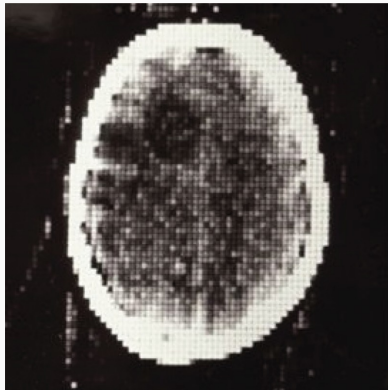
Medical Imaging: A Domain Where AI Already Matters

Why Radiology Leads

- Images are abundant, standardized, and machine-readable
- Many tasks involve classification, segmentation, and anomaly detection
- Human oversight remains essential, especially in high-stakes cases

Main Ethical Risk

If clinicians over-trust AI systems, error can become harder to detect rather than easier to prevent.



First CT scan brain image. CC BY 4.0, via Wikimedia Commons.

Risks of Medical AI

Risk	Why it matters
Bias in training data	Underrepresented groups may receive worse predictions or care
Opacity	Clinicians and patients may not understand why a recommendation was made
Overreliance	Automation bias can reduce professional vigilance
Privacy & consent	Medical datasets are intimate and difficult to fully anonymize
Access inequality	Advanced systems may benefit wealthy institutions first

WHO's central lesson: AI in health should be governed by human rights, public accountability, and the needs of health workers and patients, not just technical performance (**who_ai_health_2021**).

Diagnosis, Research, and Care Are Not the Same Problem

Medical AI is often discussed as if it were one thing. In practice, its ethical profile differs by domain.

Domain	Promise	Main risk
Diagnostics	Faster or more accurate classification	Overtrust, false confidence, bias
Research	Discovery, pattern-finding, protein structures	Unequal access, IP concentration
Administration	Lower paperwork burden, scheduling, triage	Dehumanization, hidden errors, access barriers

The strongest arguments for medical AI usually concern **augmentation**. The strongest objections concern replacing professional judgment or hiding institutional scarcity behind algorithms.

Case Study: Scarcity and Triage

Suppose a hospital uses AI to prioritize appointments, flag likely no-shows, and rank patients by risk.

Possible benefit:

- Limited staff and equipment are used more efficiently
- Some patients receive earlier intervention

Possible danger:

- The model may treat past under-treatment as evidence of lower need
- Social disadvantage can become encoded as "lower compliance" or "higher cost"
- Scarcity itself may be depoliticized: the algorithm seems objective, so no one asks why resources are insufficient

Key distinction

AI may allocate scarce care more efficiently. That is not the same thing as making the system more just.

You Be the Judge: The Hospital Board

One budget line, two proposals

Your rural hospital can fund **one**: a diagnostic imaging system that reads scans in seconds and never gets tired, or **two more radiologists**. The system is measurably good. It also fails differently than people do — and the vendor will not say exactly how.

- The system covers **every** patient, immediately, including at 3 a.m.
- The radiologists cover **fewer** patients, but can be asked why
- Your patients are, on average, not much like the population the system was trained on

Vote, then justify

Whichever you chose — name the **person** who is worse off because of it, and say what you owe them. “Nobody” is not available as an answer.

Thought Question: What Should Count as Success in Medical AI?

Discussion Question

If a medical AI system improves average accuracy but performs noticeably worse for one vulnerable group, should it still be deployed?

This question forces a choice between competing standards:

- overall utility,
- equal treatment,
- protection of the worst-off,
- or informed patient choice.

Different ethical frameworks give different answers.

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What Would Good AI Governance Mean?

Good governance means more than efficiency. Public institutions must also protect:

- Fairness and non-discrimination
- Transparency and explainability
- Contestability and due process
- Democratic legitimacy
- Privacy and civil liberties

The OECD AI Principles explicitly include **inclusive growth, human rights and democratic values, transparency, robustness, and accountability** (oecd_ai_principles_2024).

Discussion Question

Which matters more in public administration: making decisions quickly, or making them explainably and appealably? Can a government really do both well at scale?

AI in Government: Promise and Danger

Potential benefits

- Faster public-service delivery
- Fraud detection and auditing
- Translation and accessibility
- Better emergency planning

Risks

- Opaque administrative decisions
- Automated exclusion from benefits
- Surveillance and social sorting
- Reduced human accountability

OECD notes that governments are cautious adopters precisely because public decisions affect rights, not merely convenience (**oecd_ai_government_2026**).

Example

An AI translation tool in a public-benefits office might help non-English speakers access services more effectively. But an automated risk-scoring system for benefits fraud could make exclusion faster and harder to contest.

Governance Is a Political, Not Merely Technical, Problem

Why institutions matter

- AI systems used by governments should be **auditable** and **contestable**
- Public-sector automation should not replace democratic responsibility
- "Efficient" decisions can still be unjust, unappealable, or illegitimate

Lesson from Eubanks

Automation in welfare and child-protection systems often disciplines the poor rather than solving poverty
(**eubanks_automating_inequality**).



Scottish Parliament chamber. CC BY-SA 4.0, via Wikimedia Commons.

A Public-Sector Checklist

Question	Why ask it?
Can citizens appeal the decision?	If not, errors become harder to correct
Can the system be explained?	Public power should not hide behind black boxes
Is the model audited for bias?	Unequal burdens often fall on already-vulnerable groups
Is data collection proportionate?	Administrative convenience does not override rights
Does human review remain meaningful?	"Human in the loop" should not be a fiction

Efficiency logic

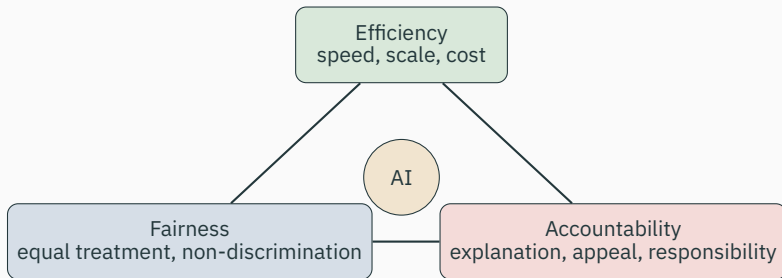
- Faster decisions
- Lower administrative cost
- More cases processed
- Uniform rule application

Legitimacy logic

- Citizens must be heard
- Reasons must be intelligible
- Errors must be reversible
- Public power must remain accountable

Democratic governance cannot be reduced to throughput. A state that decides more quickly is not necessarily a state that decides more justly.

A Governance Triangle



Public-sector AI almost always claims to improve one corner of the triangle. The political question is what gets sacrificed when efficiency is pursued without equal attention to fairness and accountability.

Why Bureaucratic AI Often Targets the Vulnerable

eubanks_automating_inequality identifies a recurring pattern:

1. Governments face scarcity, political pressure, or distrust.
2. They introduce automated screening, ranking, or fraud detection.
3. The burdens fall most heavily on those already dependent on public systems.
4. Automation then appears to solve a technical problem, while the underlying political problem remains intact.

In this sense, AI can become a technology for making exclusion look neutral.

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The Optimistic Case: AI for Development

Advocates often argue that AI can help reduce poverty by improving:

- Translation across language barriers
- Crop and weather advice for farmers
- Low-cost diagnostics
- Educational tutoring and administrative access
- Infrastructure planning and service delivery

This is not empty hype. In some contexts, AI really can extend scarce expertise or improve coordination.

Example

A mobile agricultural assistant that gives farmers localized weather alerts and crop advice may provide real value where agricultural extension services are thin. The ethical question is whether the tool strengthens local capacity or creates new dependence on outside platforms.

The Structural Critique

But development gains depend on underlying power relations.

- Who owns the models, cloud infrastructure, and chips?
- Whose languages and contexts are represented in training data?
- Which countries bear the environmental costs of extraction, energy use, and disposal?
- Who captures the profits when AI systems are deployed at scale?

sen_development_freedom reminds us that development is not merely GDP growth. It concerns real capabilities, agency, and freedom.

Discussion Question

If a country gains better AI services but loses control over its data, public institutions, and technical infrastructure, has it really become more developed?

Current Poverty and Inequality Statistics

- The World Bank estimates that about **700 million people (8.5%** of the global population) live in extreme poverty on less than \$2.15 per day (**worldbank_poverty_planet_2024**).
- About **3.5 billion people (44%** of the world) live on less than \$6.85 per day (**worldbank_poverty_planet_2024**).
- In 2024, **Sub-Saharan Africa contained 67%** of the world's extreme poor despite having only 16% of the global population (**worldbank_poverty_planet_2024**).

Implication

If AI development reproduces existing hierarchies of infrastructure, capital, and expertise, it is unlikely by itself to reduce global inequality.

What "AI for Development" Often Leaves Out

The optimistic story usually emphasizes applications. The critical story asks about preconditions.

- Reliable electricity is still absent or unstable in many places.
- Low-resource languages are underrepresented in major models.
- Cloud infrastructure and frontier compute remain concentrated in a handful of firms and countries.
- Local institutions may be asked to depend on proprietary systems they cannot inspect or govern.

This means that AI can widen dependency even when it improves some services at the margin.

From Poverty Reduction to Capability Expansion

Amartya Sen

Development is not just higher output. It is an expansion of what people are actually able to do and be.

Applied to AI, this shifts the question from:

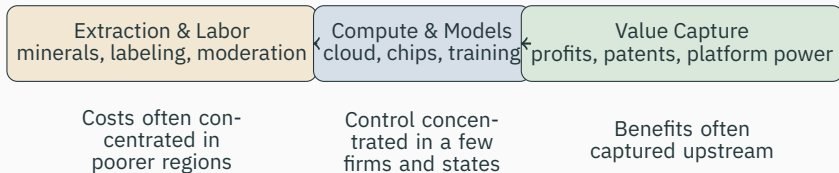
- "Does this system raise productivity?"

to:

- Does it expand human capabilities?
- Does it strengthen local agency and institutional capacity?
- Does it reduce dependency, or deepen it?

That is a better ethical test for global-development claims than raw adoption numbers.

A Global Value-Flow Diagram



This is the core concern behind talk of **extractive AI** or **AI colonialism**: unequal inputs, unequal control, unequal rewards.

Thought Question: Development for Whom?

Discussion Question

If an AI system genuinely improves service delivery in a poor country but creates long-term dependence on foreign platforms, cloud providers, and proprietary models, is that development or a new form of dependency?

The answer depends on what we think matters most:

- immediate welfare gains,
- long-term institutional autonomy,
- democratic control,
- or distributive justice.

Hidden Labor and Extraction

The material underside of AI

- Advanced computation depends on global extractive supply chains
- Mineral extraction and industrial processing are often far from where AI systems are used
- The benefits of AI tend to be concentrated, while environmental and labor costs are dispersed

Justice Question

Can an "AI for good" project still be unjust if it is built on exploitative labor or extractive infrastructures?



Artisanal cobalt miners in the DRC. CC BY 2.5, via Wikimedia Commons.

The End of the Lifecycle: E-Waste and Disposal

Why disposal matters

- Hardware has a lifecycle; replacement creates downstream waste burdens
- Recycling systems are uneven and often labor-intensive
- Disposal is another point where wealthy users can offload harms onto poorer communities

crawford_atlas_ai is useful here: AI's planetary costs do not end when the query is answered.



Electronic waste sorting. CC0, via Wikimedia Commons.

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A Framework for Evaluating AI's Social Impact

Test	Question
Sustainability	What are the energy, water, material, and waste costs?
Human benefit	Who actually gains in health, safety, knowledge, or opportunity?
Legitimacy	Is the system transparent, contestable, and democratically accountable?
Justice	Are benefits and burdens distributed fairly across groups and countries?

This framework will let us distinguish between genuinely beneficial AI applications and applications that merely shift costs onto others.

The Strongest Arguments on Each Side

Why AI may be socially valuable

- Can improve forecasting, diagnosis, and coordination
- May extend scarce expertise to underserved populations
- Can accelerate scientific discovery
- Sometimes helps address real collective-action problems

Why AI may deepen injustice

- Externalizes environmental and labor costs
- Centralizes control over data, compute, and infrastructure
- Makes exclusion and surveillance appear objective
- Risks substituting technical management for political judgment

The real task is not to choose simple optimism or pessimism. It is to identify which uses of AI are genuinely worth building, under what constraints, and for whose benefit.

Key Concepts i

Kate Crawford argues that AI is not immaterial intelligence but a material system built from extraction, labor, logistics, and planetary infrastructure (**crawford_atlas_ai**).

Rebound effect : efficiency gains can coexist with higher total resource use when cheaper systems are deployed more widely.

WHO's AI-for-health framework emphasizes human rights, public benefit, accountability, and the needs of patients and health workers (**who_ai_health_2021**).

OECD AI Principles identify inclusive growth, democratic values, transparency, robustness, and accountability as core norms for trustworthy AI (**oecd_ai_principles_2024**).

Virginia Eubanks shows that automated decision systems often manage and punish the poor rather than solving poverty (**eubanks_automating_inequality**).

Safiya Umoja Noble and **Ruha Benjamin** show that algorithmic systems can reproduce racial hierarchy while presenting themselves as neutral or objective (**noble_algorithms_oppression**; **benjamin_race_technology**).

Amartya Sen argues that development should be understood in terms of capabilities and freedom, not just aggregate output (**sen_development_freedom**).

AI colonialism / extractive AI : the pattern in which data, labor, energy, and materials are drawn from vulnerable populations while benefits and control are centralized elsewhere.

Discussion

If AI can simultaneously improve medicine, intensify resource use, and deepen inequality, what standards should determine when it ought to be built or deployed?

